

Advanced (Carolina Reaper)

Q1. Factor the expression: $x^2 + 5x + 4x + 20$

- (A) $x(x + 5) + 4(x + 5)$
- (B) $x(x + 4) + 5(x + 4)$
- (C) $x^2 + 9x + 20$
- (D) $x(x + 4) + 5(x + 4) + 4$
- (E) $x^2 + 4x + 5x + 20$

Q2. Factor the expression: $3x^2 + 12x + 4x + 16$

- (A) $3x(x + 4) + 4(x + 4)$
- (B) $3x(x + 2) + 4(x + 2)$
- (C) $3x^2 + 16x + 16$
- (D) $3x(x + 2) + 8(x + 2)$
- (E) $3x(x + 4) + 16$

Q3. Factor the expression: $x^2 + 2x + 8x + 16$

- (A) $x(x + 2) + 8(x + 2)$
- (B) $x(x + 8) + 2(x + 8)$
- (C) $x^2 + 10x + 16$
- (D) $x(x + 8) + 16$
- (E) $x(x + 2) + 16$

Q4. Factor the expression: $2x^2 + 5x + 10x + 25$

- (A) $2x(x + 5) + 5(x + 5)$
- (B) $2x(x + 2) + 10(x + 2)$
- (C) $2x^2 + 15x + 25$
- (D) $2x(x + 2) + 5(x + 5)$
- (E) $2x(x + 5) + 25$

Q5. Factor the expression: $3x^2 + 9x + 2x + 6$

- (A) $3x(x + 3) + 2(x + 3)$
- (B) $3x(x + 1) + 3(x + 1)$
- (C) $3x^2 + 11x + 6$
- (D) $3x(x + 2) + 3(x + 2)$
- (E) $3x(x + 3) + 6$

Q6. Factor the expression: $x^2 + 4x + 9x + 36$

- (A) $x(x + 4) + 9(x + 4)$
- (B) $x(x + 9) + 4(x + 9)$
- (C) $x^2 + 13x + 36$
- (D) $x(x + 9) + 36$
- (E) $x(x + 4) + 36$

Q7. Factor the expression: $2x^2 + 6x + 5x + 15$

- (A) $2x(x + 3) + 5(x + 3)$
- (B) $2x(x + 1) + 5(x + 1)$
- (C) $2x^2 + 11x + 15$
- (D) $2x(x + 5) + 3(x + 5)$
- (E) $2x(x + 3) + 15$

Q8. Factor the expression: $3x^2 + 8x + 2x + 16$

- (A) $3x(x + 2) + 8(x + 2)$
- (B) $3x(x + 8) + 2(x + 8)$
- (C) $3x^2 + 10x + 16$
- (D) $3x(x + 8) + 16$
- (E) $3x(x + 2) + 16$

Open Ended Questions (FRQ)

Q9. Factor the expression: $2x^2 + 7x + 3x + 21$. Show all steps and verify your answer.

Q10. Factor the expression: $x^2 + 6x + 4x + 24$. Show all steps and verify your answer.

Chef's Tips

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- Tip 1: Make sure to factor the GCF, if any, before factoring by grouping.
- Tip 2: Pay close attention to the signs of the terms when factoring by grouping.
- Tip 3: Verify your answers by multiplying the factors to check if the result is the original expression.